

Predictive Analytics for Customer Retention: Leveraging Machine Learning for Churn Prevention and Business Growth

Gayathri Ramasamy*

Department of Computer Science and Engineering,
Amrita School of Computing, Bengaluru,
Amrita Vishwa Vidyapeetham, India.
r_gayathri@blr.amrita.edu, ORCID ID: 0000-0003-4451-3003.

Gurupriya M

Department of Computer Science and Engineering,
Amrita School of Computing, Bengaluru,
Amrita Vishwa Vidyapeetham, India.
m_gurupriya@blr.amrita.edu.

T. Harsha Vardhan

Department of Computer Science and Engineering,
Amrita School of Computing, Bengaluru,
Amrita Vishwa Vidyapeetham, India.
bl.en.u4aie22159@bl.students.amrita.edu.

B. Sai Vivek

Department of Computer Science and Engineering,
Amrita School of Computing, Bengaluru,
Amrita Vishwa Vidyapeetham, India.
bl.en.u4aie22107@bl.students.amrita.edu.

K. Abhishek Kumar

Department of Computer Science and Engineering,
Amrita School of Computing, Bengaluru,
Amrita Vishwa Vidyapeetham, India.
bl.en.u4aie22120@bl.students.amrita.edu.

S. Kedharnath

Department of Computer Science and Engineering,
Amrita School of Computing, Bengaluru,
Amrita Vishwa Vidyapeetham, India.
bl.en.u4aie22146@bl.students.amrita.edu.

Abstract—Advanced machine learning algorithms such as Logistic Regression Random Forest and Neural Nets help businesses stop customer loss before it happens. Machine learning models track customer details and purchase behavior to detect customers who will likely stop doing business with high accuracy. With this feature companies can take planned actions that are customized for customers at risk helping them stay connected. The system helps companies use their limited resources more effectively so they save money on new customer acquisition and create happier customers. When businesses keep their current customers they build higher revenue, gain better profits and create lasting loyalty. By studying customer behavior machine learning helps businesses understand market trends ahead of time and stay ahead of the competition in today's customer-focused market. By combining predictive analytics with regular operations companies now address staff leaving while creating better ways to grow steadily.

Key Words: Machine Learning Algorithms; Predictive Analytics; Retention Strategies; Customer Churn Prediction.

I. INTRODUCTION

A customer changes service providers for business reasons, and this action is known as churning. Customers make the decision to switch to new companies. This not only results in company profits declining along with operational performance when customers choose another service provider. Our business results decline across CRM functions and operational tasks because of customer movements. CRM stands for all customer contact points under management. Establishing long-term effective customer relationships form the foundation

for institutional success. They work to bring in new clients above everything else. Service providers' customer interactions create business problems, but satisfaction needs depend on customer needs and conduct evolving expectations. Today's students need internet solutions that work differently from their predecessors. This current generation gets more education than earlier generations did. Service providers require better ways to reach their customers and tailor responsive services for them. Advanced technology understanding leads companies to restructure their work processes. The method people use to buy goods creates a big obstacle for businesses. Service providers must invent new solutions that fulfill what their customers want these expectations. Organizations have experienced success with predictive analytics approaches to retain their customers. Predictive models operated by Verizon and Vodafone in telecommunications enable these companies to deliver specific retention packages that maintain customer devotion while Netflix determines content suggestions through recorded viewing data to avoid subscription attrition. Both corporate strategies led to improved customer satisfaction results while simultaneously reducing customer turnover proportions.

II. DATASET OVERVIEW

A research-based predictive model will be used as the primary subject of study for applying it to real-time customer retention systems. The examination utilizes CRM system data which was anonymized by a telecom provider and consists

of 10,000 records containing features about call duration and service usage and subscription history as well as customer churn labels. The researchers divided the data into training and testing groups with an 80/20 ratio after cleaning and preprocessing it.

III. LITERATURE SURVEY

Companies study customer departure to seek business solutions with different machine learning methods at work across markets of interest. In research published in [1] scientists examined OTT platform customer turnover behavior. The analysis studies why people quit using Netflix or Hulu services. The tool builds predictive models based on customer profile data such as who they are, what they like and when they subscribe. Performance measurements like accuracy and recall helped us test predictions but we faced challenges from old database info and confusing data pre-processing steps. The research study suggests looking at enhanced data analysis methods and using social media information to develop more precise future estimates. In the telecom industry, another study [2]. uses machine learning techniques Logistic Regression and Decision Tree to forecast customer churn. This research examines how customers use services and pay by looking at basic personal details. The research finds that inadequate data preparation details could impact the accuracy of predictions. Future research tracks social media activity and analyzes customer sentiment to help predict customer behavior better. A related study [3] This research analyzes customer departures at Commercial Bank of Ethiopia through ANN and other predictive models. We used both customer personal details and past purchasing records in our dataset. The research produced good predictions yet ran into problems with unequal datasets as many customers never canceled their subscriptions. The study team advises making use of modern methods such as oversampling and feature modification to fix this issue with unbalanced data. Further work in the telecom sector [4] This project deals with comparable challenges by experimenting with several machine learning approaches including Logistic Regression and Decision Tree. When customers remain loyal over leaving the model cannot deliver accurate forecasts. We need to gather customer comments and track if predicted changes happen in practice to make our churn predictions better. Another study in the telecom sector [5] The study uses similar techniques with machine learning models Random Forest and ANN. The analysis points out problems with imbalanced datasets and offers solutions through customer feedback plus ongoing customer behavior tracking. E-commerce platforms create specific problems for customer retention analysis. A survey[6] The researcher looks at several studies and explains what testing procedures and measurement standards different research teams are using. Research experiences problems because it has access to scant data and faces hard task picking key features plus overlooks how buyers modify their behavior patterns. The study recommends looking at social media feedback and market data to develop better prediction tools and reveal more detailed customer perspectives. In another study

[7] We tested multiple machine learning systems including Logistic Regression, Decision Tree, Random Forest, SVM and ANN. The study finds that existing models struggle with skewed training data and need updated feature choices while lacking strong advanced algorithms. Future researchers should test ensemble methods while improving feature selection strategies and adding social media sentiment analysis to their work. A machine learning model using CatBoost[8] Our work analyzes telecom industry churn through data on customer background information and purchase habits combined with payment record details. The study separates available data into training and testing samples to test how well the model works against these standards while measuring its accuracy and correctness. To produce better results future investigations should explore modern methods and advanced machine learning algorithms. Another e-commerce-focused study [9] The study uses machine learning models to predict customer leaving patterns from their shopping and website navigation history. We measured model performance through accuracy and recall values. The writers noticed small issues with our data but recommended trying deep learning methods to make better predictions. Lastly, a study [10]. Our system combines SMOTE with machine learning to handle data imbalance when predicting customer churn. The data comes from a telecom company and shows customer information together with how they use the system. The SMOTE method successfully balanced the data which enhanced our predictive results. Ongoing research recommends developing ensemble models that collect customer sentiment data and process live data to create dynamic systems that predict churn. New technology will let churn prediction models across different businesses still work better and stay more accurate. erived from a telecom company, includes customer demographics and usage patterns. The SMOTE technique effectively balanced the dataset, leading to improved predictions. Future work recommends exploring ensemble methods, integrating customer sentiment analysis, and incorporating real-time data for dynamic churn prediction systems. These advancements could significantly enhance the adaptability and accuracy of churn prediction models across industries.

IV. PROPOSED METHODOLOGY

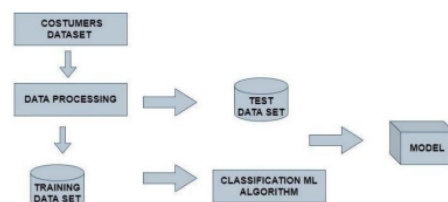


Fig. 1. Flow chart.

A. Data Collection

Our dataset came from two sources: Kaggle and the company's internal database. It contained data about our clients including their personal facts, account records, and past interactions. Our analysis begins with these datasets and uses them to make our model.

B. Data Preprocessing

To prepare data for analysis preprocessing must be performed first. Our next priority is to fix missing values using valid methods because data impacts directly on model accuracy. Machine learning algorithms require categorical data to become numeric data which one-hot encoding and quality encoding methods successfully transform. By standardizing our data characteristics we normalize numerical attributes to improve the model's processing efficiency.

C. Building the Web Application

The front-end of the platform is created using current web-engine JS-frameworks, such as React.js or Vue.js to provide a dynamic and friendly user interface. Users are able to post donation listings, browse for resources, and manage donation request all in the same space. Since the web application will be used by various users during disasters, it is compatible with multiple devices and easy to use in case of available disasters.

D. Exploratory Data Analytics(EDA)

Through EDA procedures we examine dataset types and structures while also searching for regularities and connections. The analysis uses multiple display techniques including histograms bar charts and correlations to reveal how variables work together. Our statistical methods find which aspects of the data most impact customer churn while producing important insights from the information.

E. Feature Selection

Determining the best features helps create a better churn prediction model. The correlation method helps us find strong links between target variables and potential predictor variables by preventing statistical interdependence among features. Random Forest shows how each feature influences churn rates which helps us know what details are most helpful for prediction.

F. Model Training

Our system uses machine learning technology to Identify which customers will stop using our services. Logistic Regression demonstrates promising results in binary classification tasks because of its ease of use. Random Forest gains better results by combining multiple Decision Tree models because this algorithm deals with nonlinear problems well. The algorithm supports Vector Machine helps find the ideal decision boundary while Neural Networks extracts complex data patterns.

G. Model Evaluation

We test model performance by evaluating accuracy and precision scores with F1-score plus recall numbers. These measurement tools show how well models work especially when they use datasets that are not evenly distributed. ROC-AUC enables better assessment of how well different classes stay separate during decision-making processes beyond basic performance metrics.

H. Hyperparameter Tuning

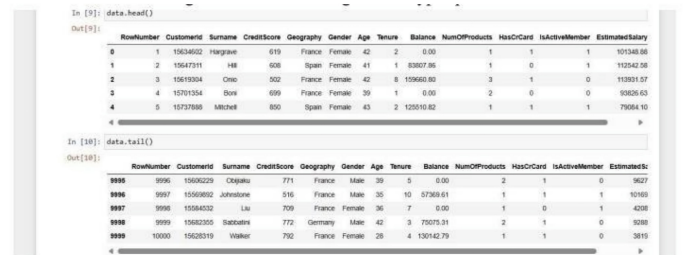
Before using the model we tune its hyperparameters to make it perform better. Both Grid Search and Random Search methods test multiple 'hyperparameters' of a model to help users find its best configuration. Measuring how each characteristic affects customer drop-out is our main focus. Our SHAP analysis shows how each input variable impacts the model's output prediction. The feature importance plots display the influence that every model feature has.

I. Deployment

Our team implements the model design and adjustments into the production system before using it to predict actual customer loss events. We move our trained model to picking and joblib formats and build a prediction API while regularly updating our model's performance.

V. RESULTS

The validation of machine learning models helps us calculate their error rate so we can measure how well they perform with our database. Small sample data sets from real-world applications need validation methods because large population-representative samples rarely require them. Validation lets us confirm if the model works correctly with real-world situations.



```
In [9]: data.head()
Out[9]:
```

	RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary
0	1	15634602	Hargrave	619	France	Female	42	2	0.00	1	1	1	101346.86
1	2	15647311	HB	635	Spain	Female	41	1	83607.86	1	0	1	112342.08
2	3	15619304	Cico	502	France	Female	42	8	159660.83	3	1	0	113391.07
3	4	15701354	Dora	689	France	Female	39	1	0.00	2	0	0	50420.43
4	5	15737888	Michel	690	Spain	Female	43	2	125510.82	1	1	1	79054.10

```
In [10]: data.tail()
Out[10]:
```

	RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary
9995	9995	15689223	Chigatu	771	France	Male	39	5	0.00	2	1	0	3627
9996	9997	15569892	Jacobson	516	France	Male	35	10	57369.63	1	1	1	10166
9997	9998	15584032	Liu	709	France	Female	36	7	0.00	1	0	1	4208
9998	9999	15682350	Stadler	772	Germany	Male	42	3	75075.31	2	1	0	5088
9999	10000	15628519	Walker	792	France	Female	28	4	130142.79	1	1	0	3815

Fig. 2. Confusion matrix highlighting key prediction results.

A portion of these data problems happen randomly. In certain situations the missing data points signal more significant problems. Understanding statistical differences in data absence is essential for our analysis. Which missing data type determines our strategy for detecting and replacing records plus full statistical methods to address missing data issues. You should first find out why data is missing to start creating programs.

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 14 columns):
#   Column              Non-Null Count  Dtype  
---  -
0   RowNumber            10000 non-null  int64  
1   CustomerId           10000 non-null  int64  
2   Surname              10000 non-null  object  
3   CreditScore           10000 non-null  int64  
4   Geography            10000 non-null  object  
5   Gender               10000 non-null  object  
6   Age                  10000 non-null  int64  
7   Tenure               10000 non-null  int64  
8   Balance              10000 non-null  float64 
9   NumOfProducts        10000 non-null  int64  
10  HasCrCard            10000 non-null  int64  
11  IsActiveMember       10000 non-null  int64  
12  EstimatedSalary      10000 non-null  float64 
13  Exited               10000 non-null  int64  
dtypes: float64(2), int64(9), object(3)
memory usage: 1.1+ MB

```

Fig. 3. Detecting Null Values and removing them.

Random Forest demonstrated 86% precision at predicting customer churn while managing difficult data sets that contain an imbalance between classes and capture indirect relationships. By linking decision trees together the method improves its prediction results which proves valuable for managing customer retention plans. Its ranking system shows us which attributes drive customer churn most strongly so businesses can enhance service quality and avoid customer loss. Finally a flask web framework has been created with the factors predicting whether the customer continues in bank or leaves.

Random Forest proved appropriate for this task due to its ability to operate at different data scale levels and provide exceptional outcomes when processing extensive datasets. The algorithm builds accuracy by combining numerous decision trees for prediction purposes while minimizing overfitting threats. An ensemble method should be used to forecast churn because it provides maximum results for analyzing mixed numerical-categorical datasets.

	precision	recall	f1-score	support
0	0.87	0.96	0.91	2379
1	0.74	0.47	0.58	621
accuracy			0.86	3000
macro avg	0.81	0.71	0.74	3000
weighted avg	0.85	0.86	0.84	3000

Fig. 4. Results of Random forest algorithm

VI. CONCLUSION

The investigation tested Random Forest along with K-Nearest Neighbors (KNN), Support Vector Machines (SVM), and Grid Search to determine customer churn predictions. Random Forest achieved the greatest accuracy of 85.6% when analyzing relationships between different variables. The accuracy ratings for KNN and SVM stood at 83% and 85%. The importance of precise predictions needs to balance with other models that are easy to interpret while running smoothly and meeting real business needs. Our team found that using

Fig. 5. Final Output.

Grid Search to test parameter combinations led to better results and enhanced performance of our model. Through our study we show business models should match with business conditions such as data organization and target achievement. Our system used customer behavior data along with economic and social media records to successfully minimize customer departure rates. Our improved method includes both basic and advanced analysis mixed with deep learning models which generate better outcomes than conventional processes while demonstrating the strength of nonlinear solutions for industry problems.

Predictive analysis delivers exact behavioral recommendations beyond the general marketing methods and broad-based incentives that were previously used. The business sector uses data-driven methods to stop potentially high-risk customers from accomplishing positive results which generate excellent outcomes.

VII. FUTURE SCOPE

The continuous process enhancement of predictive models depends on educational training approaches. Party information integrates seamlessly into training programs as organizational staff manage trends of customer behavior improvements. The performance of predictive models enhances through developer implementations of cross-validation protocols along with hyperparameter optimization techniques which include monitoring system performance. Extending the predictive solution requires an implementation of deep learning networks that include CNN and LSTM to handle time-dependent information and unstructured data. APERSE Kafka serves as an important tool to build real-time data pipelines for performing live churn prediction. Predictions will attain better accuracy through the implementation of customer sentiment analysis on social media content.

REFERENCES

- [1] O. R. Devi, S. K. Pothini, M. P. Kumari, S. V., and U. N. S. Charan, "Customer Churn Prediction using Machine Learning: Subscription Renewal on OTT Platforms," *2023 2nd International Conference on Applied Artificial Intelligence and Computing (ICAAC)*, Salem, India, 2023.

- [2] R. Srinivasan, D. Rajeswari, and G. Elangovan, "Customer Churn Prediction Using Machine Learning Approaches," *2023 International Conference on Artificial Intelligence and Knowledge Discovery in Concurrent Engineering (ICECONF)*, Chennai, India, 2023.
- [3] M. H. Seid and M. M. Woldeyohannis, "Customer Churn Prediction Using Machine Learning: Commercial Bank of Ethiopia," *2022 International Conference on Information and Communication Technology for Development for Africa (ICT4DA)*, Bahir Dar, Ethiopia, 2022.
- [4] V. Agarwal, S. Taware, S. A. Yadav, D. Gangodkar, A. Rao, and V. K. Srivastav, "Customer Churn Prediction Using Machine Learning," *2022 2nd International Conference on Technological Advancements in Computational Sciences (ICTACS)*, Tashkent, Uzbekistan, 2022.
- [5] R. K. Peddarapu, S. Ameena, S. Yashaswini, N. Shreshta, and M. Purnasahithi, "Customer Churn Prediction using Machine Learning," *2022 6th International Conference on Electronics, Communication and Aerospace Technology*, Coimbatore, India, 2022.
- [6] P. Gopal and N. B. Mohd Naw, "A Survey on Customer Churn Prediction using Machine Learning and Data Mining Techniques in E-commerce," *2021 IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE)*, Brisbane, Australia, 2021.
- [7] H. Karamollaoğlu, İ. Yücedağ, and İ. A. Doğru, "Customer Churn Prediction Using Machine Learning Methods: A Comparative Analysis," *2021 6th International Conference on Computer Science and Engineering (UBMK)*, Ankara, Turkey, 2021.
- [8] J. Jane Rubel Angelina, S. J. Subhashini, S. Harish Baba, P. Dheeraj Kumar Reddy, P. V. Sudheer Kumar Reddy, and K. Sameer Khan, "A Machine Learning Model for Customer Churn Prediction using CatBoost Classifier," *2023 7th International Conference on Intelligent Computing and Control Systems (ICICCS)*, Madurai, India, 2023.
- [9] P. Nagaraj, V. Muneeswaran, A. Dharanidharan, M. Aakash, K. Balanathanan, and C. Rajkumar, "E-Commerce Customer Churn Prediction Scheme Based on Customer Behaviour Using Machine Learning," *2023 International Conference on Computer Communication and Informatics (ICCCI)*, Coimbatore, India, 2023.
- [10] A. O. Akinrotimi, R. O. Ogundokun, M. A. Mabayoje, R. A. Oyekunle, and M. O. Adebisi, "A SMOTE-Based Churn Prediction System Using Machine Learning Techniques," *2023 International Conference on Science, Engineering and Business for Sustainable Development Goals (SEB-SDG)*, Omu-Aran, Nigeria, 2023.
- [11] Rohit, Y. S. S., T. Rushik, and Gayathri Ramasamy. "A Comparative Analysis of Machine Learning Models for Stock Price Prediction Performance Evaluation and Improvement Strategiesve Analysis of Machine Learning Models for Stock Price Prediction." Available at SSRN 5089161 (2024).
- [12] Bindu Sree, Mandapati, et al. "Prediction of Risk of Coronary Heart Disease Using Various Machine Learning Models." Dasari Keerthi Sai and Sirishma, Amara Gnana and Kumar Reddy, Patil Hemanth and Reddy Ys, Prem and Ramasamy, Gayathri, Prediction of Risk of Coronary Heart Disease Using Various Machine Learning Models (January 9, 2025) (2025).
- [13] Sirishma, Amara Gnana, et al. "Machine Learning For Risk Assessment: A Comparative Study of Models Predicting Loan Approval." Machine Learning For Risk Assessment: A Comparative Study of Models Predicting Loan Approval (November 15, 2024) (2024).
- [14] Gayathri, R., et al. "A framework for the prediction of diabetes mellitus using hyper-parameter tuned xgboost classifier." *2022 13th International Conference on Computing Communication and Networking Technologies (ICCCNT)*. IEEE, 2022.
- [15] Krishna, Kaivaram Sivarama, et al. "A multi-class classification framework with smote based data augmentation technique for Alzheimer's disease progression." *2024 International Conference on Advances in Modern Age Technologies for Health and Engineering Science (AM-ATHE)*. IEEE, 2024.